

NORTH CORMORANT

DCS
MAIN
OPERATOR
POINT

DCS

CONTROL &
ALARM
SIGNALS

DATA AND
COMMUNICATION

MCS

PCU

MC

OS

RTDS

PRINTER

CONF./
HIST. SERVER

EWS/
WEB SERVER

EPU
POWER
MODEM

POWER

ETU/PC FOR
MFM

UPS

POWER

EXISTING
MCC

HPU

ESD

EXISTING
ESD

2 x HP LINES
2 x LP LINES

METHANOL
INJECTION

CHEMICAL
INJECTION

KHI INJ.

TUTU

ACCESS FOR
ONSHORE DIAGNOSTIC
(READ ONLY)

SIGNAL ON POWER COMMUNICATION

DCS

EXISTING ESD

VSD

ESP CONTR.
DATA MONITORING

JB

WATER DEPTH:
184 m

MAIN UMBILICAL
21 km TO OTTER FIELD
REF. PAGE 2,3 AND 4.

ELECTRIC
POWER CABLES
TO ESP SYSTEM.

DELIVERED
BY OTHERS

LEGEND



MQC-INBOARD - MQC-12 / PARKING



MQC, MK-II OUTBOARD



MQC, PROTECTION CAP



4-PIN / 12-PIN ELECTRIC
CONNECTOR / PROTECTION



4-PIN / 12-PIN MALE PARKING



EL. CONNECTION PAIR, DIVER OPERATED



WEAK LINK

LEGEND:

MFM: MULTIPHASE METER

SCM: SUBSEA CONTROL MODULE

SUTU: SUBSEA UMBILICAL TERMINATION UNIT

WI: WATER INJECTION

PROD: PRODUCTION

TEST EQUIPMENT NOTE 8

SCM
DECK STAND
(SHIPPING
FRAME)

TEST &
FLUSHING
PLATES

SCM TEST
STAND

POS	DESCRIPTION	QTY
1	HYDRAULIC POWER UNIT (HPU)	1
2	MASTER CONTROL STATION (MCS)	1
3	ELECTRIC POWER UNIT (EPU)	1
4	TOPSIDE UMBILICAL TERMINATION UNIT (TUTU)	1
5	N/A	-
6	ETU/PC FOR MFM	1
7	UMBILICAL, 21 km	1
8	N/A	-
9	N/A	-
10	SCM DECK STAND (SHIPPING FRAME)	1
11	N/A	-
12	SCM TEST STAND (SCMTS)	1
13	TEST & FLUSHING PLATES (2 x OUTBOARD, 1 x INBOARD)	SET
14	N/A	-
15	N/A	-
16	SUBSEA UMBILICAL TERMINATION UNIT (SUTU)	1
17	EL. JUMPER 1-7, P/S	2
18	EL. JUMPER 1-1 P/S	2
19	MFM HUB	-
20	MULTIPHASE FLOW METER (MFM), ROXAR	1
21	N/A	-
22	4-PIN MALE ROV, PARKING	16
23	EL. JUMPER 1-3, 1 x PT, 1 x PT/TT, 1 x PD	1
24	FEMALE ROV, LTP	21
25	MQC 8 LINES, INBOARD	2
26	HYDR. JUMPER, 8 LINES, 50 m W/MQC	1
27	POWER SWITCH	1
28	2 L HYDR. ROV COUPLER, FEMALE	1
29	HYD/CHEM. JUMPER, 13+1 LINES, 10 m	1
30	N/A	-
31	2 L HYDR. ROV COUPLER, MALE PARKING	1
32	WEAK LINK	1
33	HYDRAULIC SINGLE RECEPTACLE	4
34	MQC-13, OUTBOARD PROTECTION CAP	3
35	4-PIN HV MALE PARKING	2
36	NA	-
37	MQC-13 LINES, INBOARD	1
38	MQC-13, PARKING	1
39	SUBSEA ACCUMULATOR	2
40	4-pin Female ROV Protection	6
41	SCM (18 x LP, 1 x HP)	5+1S
42	SCMMB (18 x LP, 1 x HP)	5+1S
43	EL. JUMPER 1-2, 1 x PT/TT, 1 x PT, ST. ACCURACY, RS, ROV	1
44	EL. JUMPER 1-2, 2 x PT/TT, ST. ACCURACY, RS	6+2S
45	EL. JUMPER 1-2, 2 x PT, ST. ACCURACY, RS	4
46	EL. JUMPER 1-2, PT/TT, LVDT, ASD	3+1S
47	NA	-
48	EL. JUMPER 1-1, LVDT	2

NOTE: SCOPE OF SUPPLY INCLUDES 1 OFF SPARE PRODUCTION XT INCL. SCM
S=SPARE
POWER CABLE TERMINATION IS NOT FMC SCOPE

NOTES:

- NOT SCOPE OF SUPPLY.
- EWS: ENGINEERING WORK STATION.
- OS: OPERATOR STATION.
- TOPSIDE CABLING AND PIPING ARE NOT PART OF SCOPE.
- E = ELECTRICAL.
- ADVANCED DATA MODULE FOR THE MPFM (MFM) IS NOT INCLUDED.
- ASD (CLAMP ON) CALCULATING, SAND-RATE BASED ON PRESSURE DIFFERENTIAL MEASUREMENT ACROSS XT CHOKE.
- MCS/SCM SIMULATOR FUNCTIONS INCLUDED IN ETU FOR MFM.

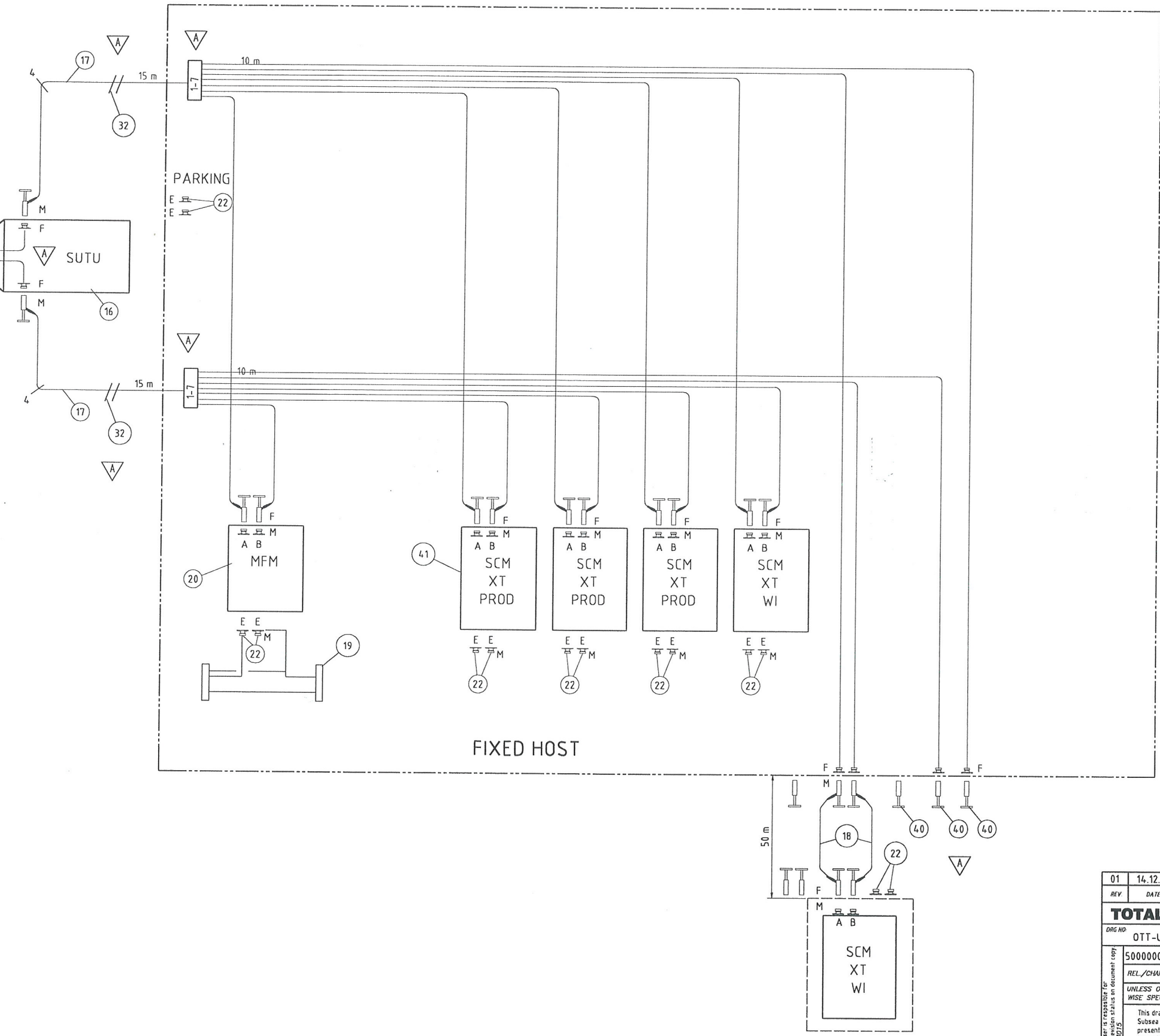
no off	SERVICE	INJECTION POINT	PRESSURE (BAR)	LINE SIZE
1	WAX INHIBITOR	MANIFOLD	345	12.7 mm
1	DEMULSIFIER	MANIFOLD	345	12.7 mm
4	SCALE INHIBITOR (1 SPARE)	XT	210	9.5 mm
1	CORROSION INHIBITOR	MANIFOLD	345	12.7 mm
1	H ₂ S SCAVENGER	MANIFOLD	345	12.7 mm
2	METHANOL	XT	345	25.4 mm
2	KINETIC HYDRATE INHIBITOR	MANIFOLD	345	12.7 mm
2	LP HYDRAULIC		230	12.7 mm
2	HP HYDRAULIC		517	9.5 mm
2	SPARE LINE		345	12.7 mm
2	EL. QUADS			6 mm2
2	SPARE LINES		345	9.5 mm

OPERATIONAL PRESSURE: HP:517 bar, LP:207 bar
CONTROL FLUID: MARSTON BENTLEY HW540

RECEIVER: *Client*
AS BUILT
02 JAN. 2002
SIGN. *Ab*
KONGSBERG OFFSHORE a.s.

01	14.12.01	AS BUILT	BKO	LAN	JTT						
REV	DATE	DESCRIPTION	DRN	DRG	ENG	APP	PROJ	APP	TRF	APP	
TOTAL FINA ELF											
OTTER FIELD DEVELOPMENT											
DRG NO: OTT-U10-PT-80-1003											
Contractor: Berit Kopseng											
SHT. 01											
REV. 01											
500000009529											
DESIGNER: Berit Kopseng											
LAST SAVED (DATE/TIME): 20.12.2001 15:21											
NO. OF REV. 15											
UNLESS OTHERWISE SPECIFIED											
DIM. TOL. ROUGHNESS R _a μm: 3.2											
Fillet RAD. BROKEN EDGES (R ALT. 45°/60° - 90°)											
THREAD TOLERANCE: 6g/6h ISO 1914 - ISO 1915/1											
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SCALE 1:1											
A1											
FMC Energy Systems											
FMC Kongsberg Subsea											
SYSTEM DRAWING											
TOPSIDE											
TOTALFINA - OTTER DEVELOPMENT											
DRAWING NO. DWG-0228155-001											
WEIGHT: SHEET 1 OF 4											

7
MAIN
UMBILICAL
21 km



RECEIVER: *Client*
AS BUILT
02 JAN. 2002
SIGN. *At*
KONGSBERG OFFSHORE a.s.

0114.12.01AS BUILT

BKO

LAN

JTT

REVDATEDESCRIPTION

DRN

DRGCHK

ENGCHK

ENGAPP

PROJAPP

TECHAPP

TOTAL FINA ELF

OTTER FIELD DEVELOPMENT

DRG NO:OTT-U10-PT-80-1003

Contractor

SHT.02

REV.01

500000009529Berit Kopseng20.12.200115:106A

REL/CHANGE NO.DESIGNERLAST SAVED (DATE/TIME)NO. OF REV.

UNLESS OTHER-WISE SPECIFIEDDIM. TOL.ROUGHNESS Ra µmFILLET RAD.BROKEN EDGES (R ALT. 45°/1.1 - 0.5)THREAD TOLERANCE: 6g/6h (NS 1014 - ISO R165/1)

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SCALE1:1

SIZEA1

SYSTEM DRAWINGDRAWING NO. DWG-0228155-002

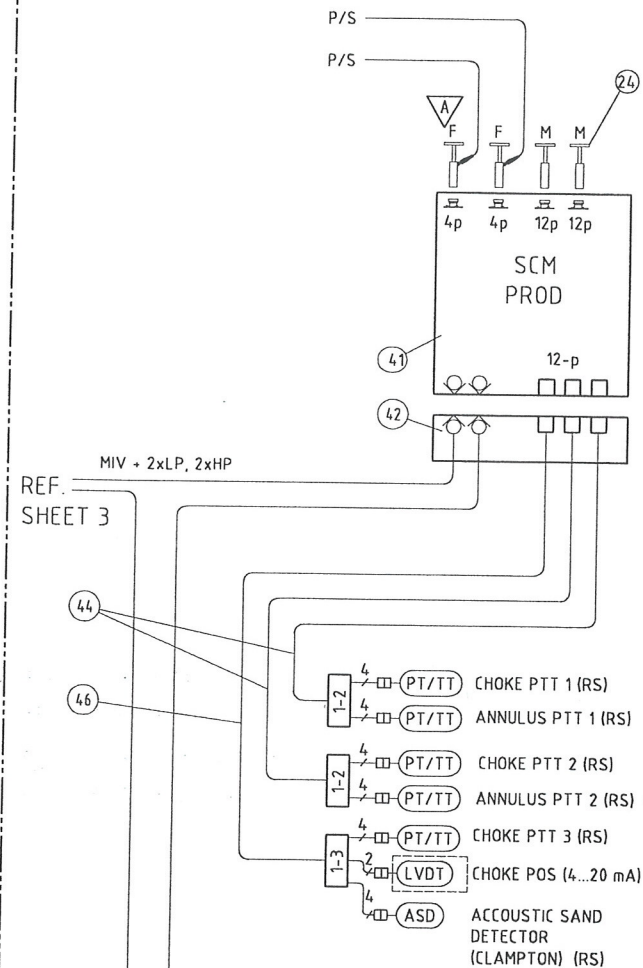
EL. DISTRIBUTION

TOTALFINA - OTTER DEVELOPMENTWEIGHT:SHEET 2 OF 4

101112

ELECTRONIC ORIGINAL: User is responsible for verification of current revision status on document copy. TOLERANCING ISO 8015

PRODUCTION XT (3 OFF)



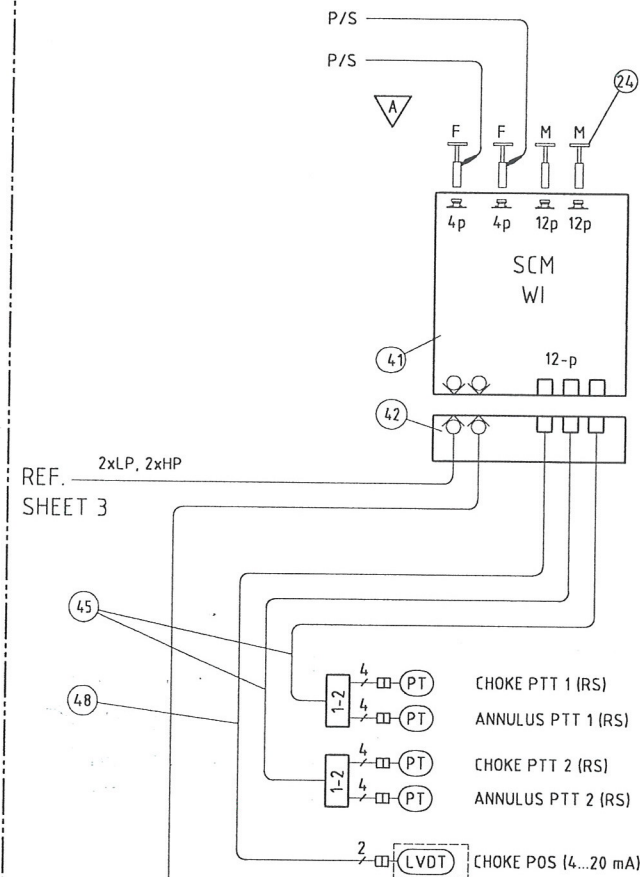
LP: 18 LINES, HP: 1 LINE

HYD. FUNCTIONS ON PROD. XT

NO.	FUNCTION	UNIT	HP	BAR
53	SCSSV	HP	517	bar
3	PWV	LP	207	bar
4	PMV	LP	207	bar
5	XOV	LP	207	bar
6	AMV	LP	207	bar
7	MIV	LP	207	bar
8	SIV	LP	207	bar
9	CV (O)	LP	207	bar
10	CV (C)	LP	207	bar
11	FSP SWITCH UNIT	LP	207	bar
13	ESP SWITCH UNIT	LP	207	bar
12	PRV	LP	207	bar
14	HYD. CONTR.	LP	207	bar
15	HYD. CONTR.	LP	207	bar
16	HYD. CONTR.	LP	207	bar
17	HYD. CONTR.	LP	207	bar
18	HYD. CONTR.	LP	207	bar
19	HYD. CONTR.	LP	207	bar
20	SPARE	LP	207	bar

CHEMICAL INJ.
SCALE
METHANOL

WATER INJECTION XT (2 OFF)



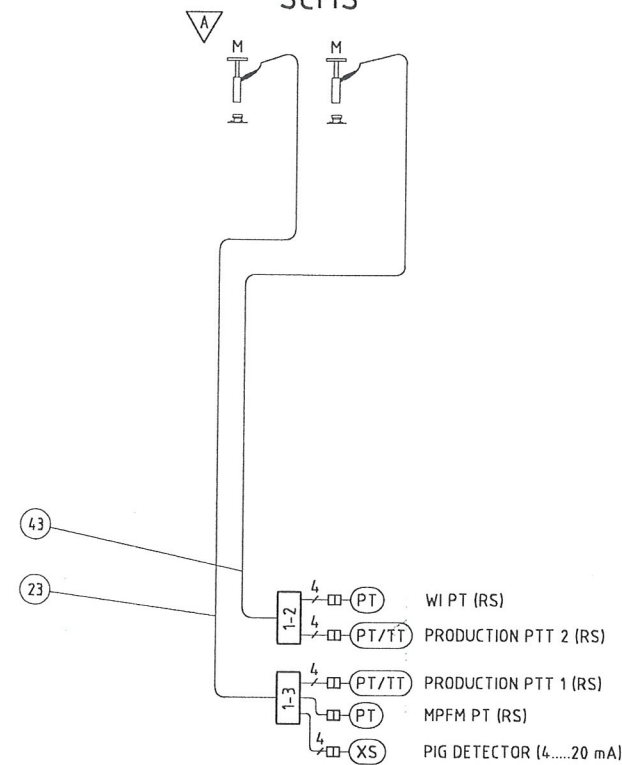
LP: 18 LINES, HP: 1 LINE

HYD. FUNCTIONS ON WI XT

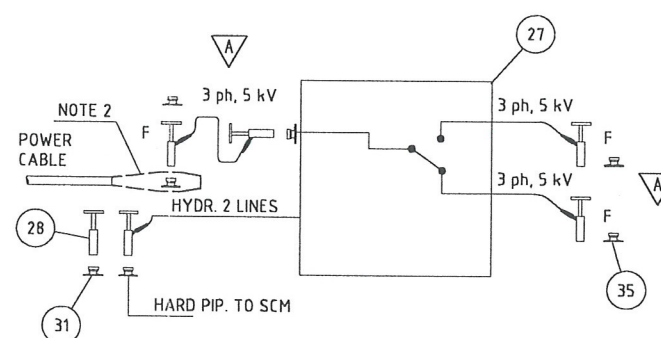
NO.	FUNCTION	UNIT	HP	BAR
53	SCSSV	HP	517	bar
3	WIWV	LP	207	bar
4	WIMV	LP	207	bar
5	XOV	LP	207	bar
6	AMV	LP	207	bar
7	SPARE	LP	207	bar
8	SPARE	LP	207	bar
9	CV (O)	LP	207	bar
10	CV (C)	LP	207	bar
11	SPARE	LP	207	bar
12	SPARE	LP	207	bar
13	SPARE	LP	207	bar
14	SPARE	LP	207	bar
15	PIV1(O)	LP	207	bar
16	PIV1(C)	LP	207	bar
17	PIV2(C)	LP	207	bar
18	PIV2(O)	LP	207	bar
19	SPARE	LP	207	bar
20	SPARE	LP	207	bar

MANIFOLD / PLEM

SENSORS TO BE CONNECTED TO ONE OF THE SCMS



POWER DISTRIBUTION



RECEIVER: *Client*
AS BUILT
02 JAN. 2002
SIGN: *Abh*
KONGSBERG OFFSHORE a.s.

REV.	DATE	DESCRIPTION	DRN	ENG	APP	PROJ	TECHN
01	14.12.01	AS BUILT	BKO	LAN	JTT		

TOTAL FINA ELF

DRG NO: OTT-U10-PT-80-1003

Contractor: *Client*

SHIT: 04

REV: 01

500000009529 KOH 14.12.2001 10:52 9 A

REL/CHANGE NO. DESIGNER LAST SAVED (DATE/TIME) NO. OF REV.

UNLESS OTHERWISE SPECIFIED DIM. TOL. ROUGHNESS Ra µm: FILLET RAD. BROKEN EDGES (R ALT. 45°) 0.5 - 0.5 THREAD TOLERANCE: 6g/6h ISO 1814 - ISO R15/14

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SCALE: 1:1

SIZE: A1

SYSTEM DRAWING

SCM FUNCTIONS

TOTALFINA - OTTER DEVELOPMENT

DRAWING NO. DWG-0228155-004

WEIGHT: SHEET 4 OF 4